

AAFA-Net: Adaptive Amplitude-Frequency Attention Network for Fetal Monitoring From Noninvasive Abdominal Recordings

Yong Liao¹, Xu Wang¹, Yang Han¹, and Yamei Deng¹

Abstract—The noninvasive fetal electrocardiogram (FECG) is helpful for fetal well-being monitoring. However, it is difficult to obtain high-quality FECG signals because of the maternal electrocardiogram (MECG) and noise in the abdominal ECG (AECG). To address this problem, an adaptive amplitude-frequency attention network (AAFA-Net) is proposed for extracting FECG signals from AECG signals, where the frequency encoder-decoder (FED) module is developed to distinguish the FECG frequency components from AECG signals, and the amplitude encoder-decoder (AED) module is devised to extract FECG amplitude components from AECG signals, while the window encoder-decoder (WED) module is designed to determine the temporal window around the FECG signal. Experiments conducted on the benchmarks show that the proposed AAFA-Net performs better than the state-of-the-art approaches.

Index Terms—Attention mechanism, fetal electrocardiogram (FECG), Fourier series, frequency-amplitude.

I. INTRODUCTION

RECENTLY, electronic monitoring devices play an important role in fetal monitoring [1], [2], e.g. fetal magnetocardiography (FMCG) [3] and Doppler ultrasound [4], [5]. They can not only monitor the development of the fetus but also detect some diseases, e.g., congenital heart disease. Although Doppler ultrasound is widely used in fetal health monitoring, it is only safe within the exposure threshold [5]. Besides, it is too expensive to monitor fetal heart rate through FMCG [6]. Therefore, it is essential to design a

safe and affordable tool to diagnose fetal well-being during pregnancy.

Fetal electrocardiogram (FECG), one of the effective tools for fetal diagnosis, has obtained much attention due to its ability to provide effective information on fetal development [7]. For example, the fetal heart rate is related to oxygen deprivation [8]. Despite the strong capability of representing fetal information, FECG signals always suffer from noise, such as power-line frequency interference, EMG signal, and high-amplitude maternal ECG (MECG) signal [9], [10], resulting in a failure of fetal diagnosis.

To solve this problem, researchers have proposed many methods to separate FECG signals from abdominal ECG (AECG) signals [11], [12], [13], [14]. For example, Mohebian et al. [15] propose a cycle generative adversarial neural network (CycleGAN) to map MECG to FECG using the attention mechanism. However, some MECG signals are wrongly predicted as FECG signals by CycleGAN due to their high-amplitude waveform. In addition, Warmerdam et al. [8] design a hierarchical probabilistic framework to detect fetal *R*-Peak from the low signal-to-noise AECG signals by combining ECG waveform and heart rate information, but it is limited to detect the fetal *R* peaks and cannot perform ECG waveform analysis. Besides, Andreotti et al. [16] exploit a naive Bayes classifier and an adaptive Kalman filter to evaluate the multi-channel fetal ECG signal from various noise sources, but only for fetal heart rate prediction. Zhang et al. [17] develop a fixed-point blind source separation algorithm to separate the FECG signal from the AECG signal corrupted by the noise, where the generalized autocorrelation and the nonlinear predictability are utilized to improve the algorithm performance, but it is only validated in one limited FECG dataset. In short, these methods detect FECG signals from AECG signals by feature extraction.

However, various noise sources still affect the quality of the abdominal signals, resulting in a challenging task for FECG signal detection. First, the FECG signal is always corrupted by the noise, leading to waveform distortion, further adversely affecting the FECG signal quality, as shown in the red circle of Fig. 1(a). Second, most FECG signals are masked by high-amplitude MECG signals [see the red box in Fig. 1(a)], and these FECG signals are likely to be regarded as noise by networks. Third, FECG signals overlap with MECG signals in the time and frequency domain [see Fig. 1(b)], always resulting in residual MECG components in the FECG signal.

Manuscript received 27 February 2023; revised 6 September 2023; accepted 29 September 2023. Date of publication 20 October 2023; date of current version 31 October 2023. This work was supported by the National Natural Science Foundation of China under Grant 51975129 and Grant 6197020346. The Associate Editor coordinating the review process was Dr. Bruno Andò. (Corresponding author: Xu Wang.)

Yong Liao is with the Microelectronics and Optoelectronics Technology Key Laboratory of Hunan Higher Education, School of Physics and Electronic Electrical Engineering, Xiangnan University, Chenzhou 423000, China (e-mail: liaoyongxnxxy@gmail.com).

Xu Wang is with the School of Automation, Guangdong University of Technology, Guangzhou 510006, China (e-mail: xuwang.scholar@gmail.com).

Yang Han is with the Department of Obstetrics, The First Affiliated Hospital, Sun Yat-sen University, Guangzhou 510080, China (e-mail: hanyang@mail2.sysu.edu.cn).

Yamei Deng is with the Department of Radiology, Guangdong Provincial Key Laboratory of Major Obstetric Diseases, Guangdong Provincial Clinical Research Center for Obstetrics and Gynecology, The Third Affiliated Hospital of Guangzhou Medical University, Guangzhou 510150, China (e-mail: dengyamei0726@gmail.com).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TIM.2023.3326161>, provided by the authors.

Digital Object Identifier 10.1109/TIM.2023.3326161

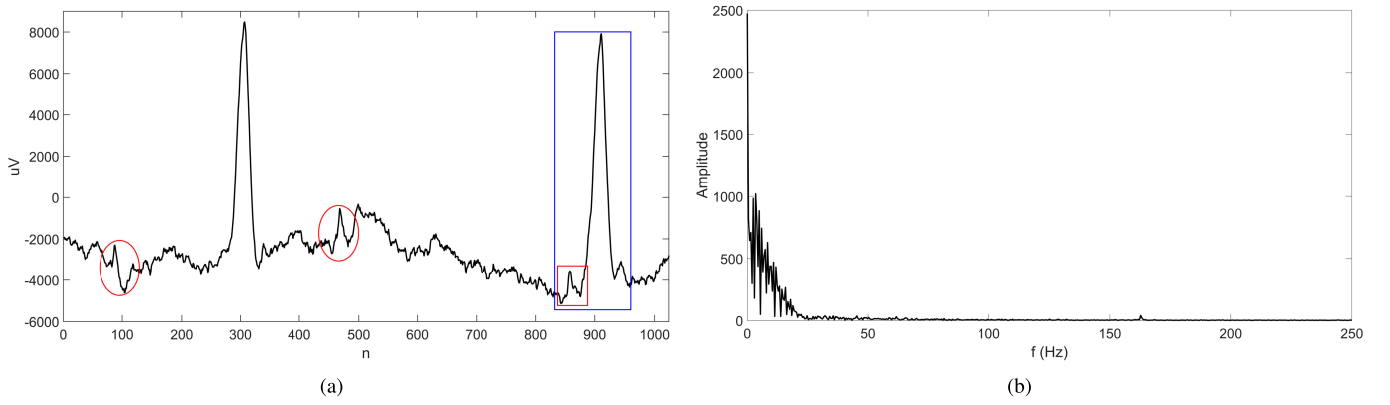


Fig. 1. Examples of the FECG signals corrupted by the MECG signal and noise. (a) Time domain. (b) Frequency domain.

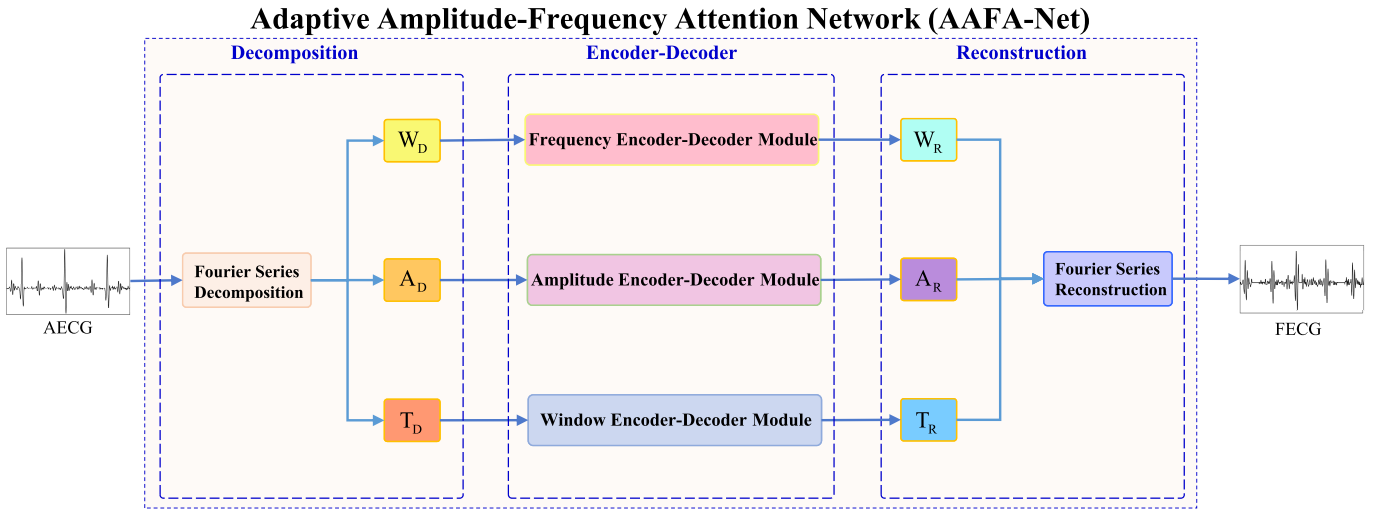


Fig. 2. Architecture of AAFA-Net. The AECG signals are first transformed into three feature maps, i.e., frequency W_D , amplitude A_D , and window T_D after FSD. Then, the frequency feature maps W_D are fed into the FED module to distinguish FECG frequency components W_R from MECG signals, while the amplitude feature maps A_D are transmitted into the AED module to obtain FECG amplitude components A_R masked by MECG signals, and the window feature maps T_D are fed into the WED module to determine the temporal windows T_R of FECG signals corrupted by noise. Finally, these three feature spaces are utilized to generate the FECG signals through FSR.

To address such problems, an adaptive amplitude-frequency attention network (AAFA-Net) is proposed for FECG signal extraction, where the Fourier series decomposition (FSD) is first applied to transform AECG signals into the frequency, amplitude, and window spaces; then, the frequency encoder-decoder (FED) module is devised to distinguish the FECG signals from the MECG signals by solving the FECG frequency components, and the amplitude encoder-decoder (AED) module is designed to extract the FECG components masked by MECG signals by obtaining the FECG amplitude components, while the window encoder-decoder (WED) module is developed to separate the FECG signals corrupted by noise by determining the temporal window around FECG signals. Finally, the Fourier series reconstruction (FSR) is exploited to reconstruct the FECG signals based on the frequency, amplitude, and window spaces of FECG signals. The main contributions are listed as follows.

1) The FED module, which can solve the FECG frequency components from the AECG frequency space, is devised to distinguish FECG signal components from MECG signals.

- 2) The AED module, which can obtain the FECG amplitude components from the AECG amplitude space, is designed to extract the FECG components masked by MECG signals.
- 3) The WED module, which can determine the temporal window around FECG signals in the sequence, is developed to separate the FECG signals corrupted by noise.
- 4) Extensive experiments conducted on the B2_LABOUR and ADFECGDB datasets validate the effectiveness of the proposed AAFA-Net for FECG extraction.

The remainder of this article is shown as follows. Section II presents the proposed AAFA-Net. Materials and evaluation metrics are shown in Section III. Section IV describes the experiments and results analysis. Section V shows the discussions, while conclusions are given in Section VI.

II. AAFA-NET FOR FETAL MONITORING

In this section, an AAFA-Net is proposed for FECG separation, as shown in Fig. 2. It consists of five parts: FSD,

TABLE I

CONFIGURATION OF THE PROPOSED FED, AED, AND WED MODULES. "LSTM," "CONV1D," "LINEAR," "MHA," AND "MATRIX" DENOTE THE LONG-SHORT TERM MEMORY NETWORK, 1-D CONVOLUTION, LINEAR MATRIX, MULTI-HEAD ATTENTION, AND RANDOM INITIALIZED MATRIX, RESPECTIVELY

Module	Layer Name	Configuration
FED	LSTM	$[77 \times 78] \times 1$
	LSTM	$[77 \times 78] \times 1$
	Conv1d	$[38 \times 38, 1] \times 1$
AED	Conv1d	$[1 \times 1, 1] \times 3$
	Linear	$[77 \times 77] \times 3$
	MHA	$[77 \times 77] \times 2$
	Linear	$[77 \times 77] \times 3$
	Conv1d	$[1 \times 1, 1] \times 3$
	Conv1d	$[1 \times 1, 3] \times 1$
WED	Matrix	$[38 \times 128] \times 1$
	Linear	$[128 \times 128] \times 1$
	LSTM	$[128 \times 128] \times 1$
	Conv1d	$[3 \times 3, 38] \times 1$
	Linear	$[128 \times 128] \times 1$

FED module, AED module, WED module, and FSR, and the configuration is given in Table I, which are described in detail as follows.

A. Fourier Series Decomposition

In general, the Fourier series [18], [19] can be expressed as

$$H(t) = \sum_{n=1}^{\infty} [a_n \sin(nwt)] \quad (1)$$

where a_n is the amplitude of signals, w denotes the resolution frequency under-sampling, and n represents the resolution depth. After applying (1) to the AECG (A) signals, we have a collection of frequencies a_n^A and amplitudes nw^A

$$A = \{[a_1^A, \dots, a_n^A], [w^A, \dots, nw^A]\}. \quad (2)$$

Then, we introduce a window function to determine the coordinates of FECG signals. Thus, (2) can further be expanded as

$$A = \begin{Bmatrix} [a_1^A, \dots, a_n^A] \\ [w^A, \dots, nw^A] \\ [u_1^A, \dots, u_n^A] \end{Bmatrix} \quad (3)$$

where u_n denotes the window function, described as

$$u = T(t - w). \quad (4)$$

Based on it, the formula of FECG separation can be defined as

$$F(t) = \sum_{n=1}^{\infty} T_F(t - w) [A_n^F \sin(nwt)]. \quad (5)$$

Therefore, the map between FECG and AECG signals can be described as

$$A = \begin{Bmatrix} [a_1^A, \dots, a_n^A] & \dashrightarrow & [a_1^F, \dots, a_n^F] \\ [w^A, \dots, nw^A] & \dashrightarrow & [w^F, \dots, nw^F] \\ [u_1^A, \dots, u_n^A] & \dashrightarrow & [u_1^F, \dots, u_n^F] \end{Bmatrix} = F. \quad (6)$$

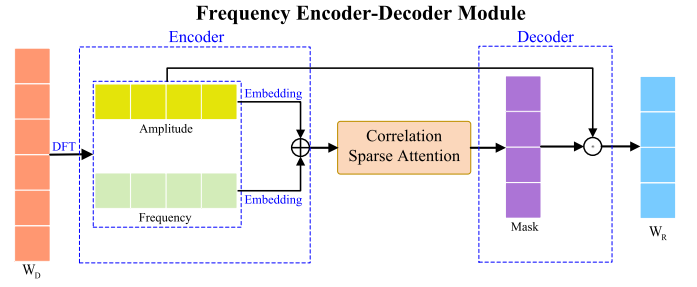


Fig. 3. Structure of FED module. The frequency feature maps W_D are first converted into amplitude and frequency components after DFT. Then, the correlation sparse attention is exploited to generate the frequency mask, which will be multiplied with frequencies to obtain the FECG frequency components, where $\oplus$ and $\odot$ represent the matrix addition and element-wise Hadamard product, respectively.

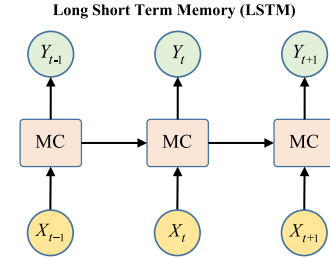


Fig. 4. Structure of the LSTM network, where MC denotes the memory cell.

It can be seen from (6) that the FECG separation can be regarded as resolving the frequency, amplitude, and window of the FECG signal from AECG signals. However, the traditional FSD algorithms are time-consuming. To this end, three basic convolution layers, i.e., Conv1d, are employed to produce different three feature maps, which include frequency feature map W_D , amplitude feature map A_D , and window feature map T_D . Next, the frequency, amplitude, and WED modules are described in detail.

B. FED Module for Solving FECG Frequency Components

The FECG and MECG signals are often mixed in the AECG signals, and also aliased in the frequency domain, resulting in a challenging task for FECG signal separation. To solve this problem, a FED module is designed to decompose the frequency of FECG signals from AECG signals, where the long short term memory (LSTM) network is utilized to learn the dependencies between amplitude and frequency corresponding to each coordinate in the sequence, as shown in Fig. 3, which is described as follows:

- 1) First, the frequency feature maps W_D are converted into amplitude and frequency components after the discrete Fourier transform (DFT).
- 2) Then, the amplitude and frequency components are processed by embedding and added to generate word embedding vectors, where embedding is implemented by the LSTM network, as shown in Fig. 4. In the LSTM network, the current memory cell not only generates output information, but also transfers it to the next one, effectively achieving sequence modeling and long-term dependency learning.

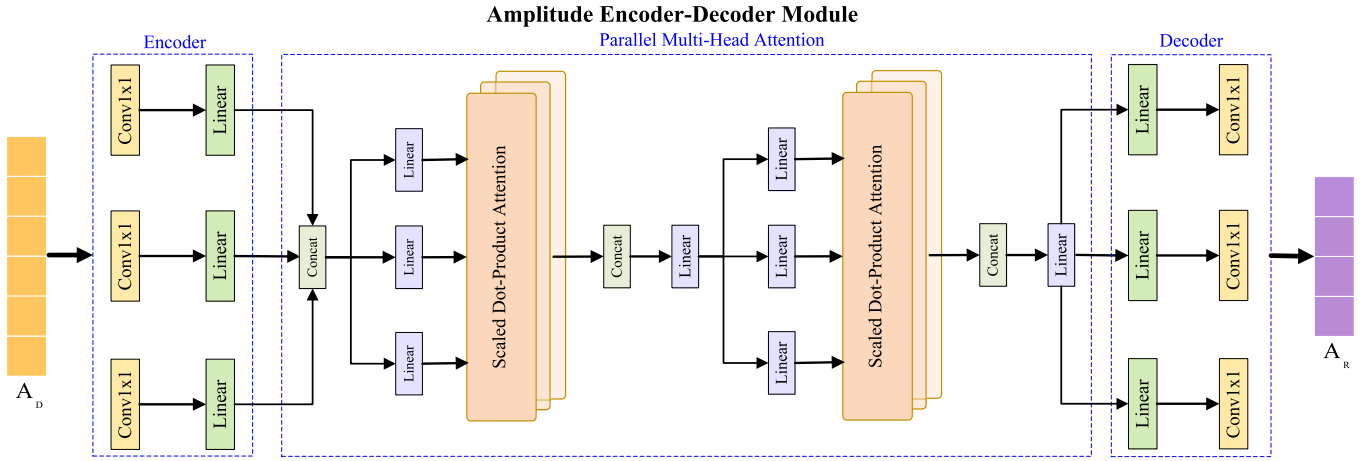


Fig. 5. Structure of AED module. First, the encoder, consisting of conv2d layers and linear layers, is used to generate three feature spaces: query, key, and value, respectively. Then, these feature spaces are fed into the parallel multi-head attention [21] to extract the amplitude components of FECG signals. Finally, the decoder, which contains linear and conv1d layers, is applied to produce the FECG amplitude components.

- 3) Next, the correlation sparse attention is applied to produce a top- n frequency mask, where sparse attention comes from Informer [20], defined as

$$\text{SparseAtt}(Q, K, V) = \text{Softmax}\left(\frac{\tilde{Q}K^T}{\sqrt{d}}\right)V \quad (7)$$

where Softmax denotes the softmax function, which is used to normalize the weights and highlight the important elements, $\tilde{Q}$ denotes the Top- n queries through the sparsity measurement $\tilde{M}(q, K)$ conducted on the original queries Q

$$\tilde{M}(q_i, K) = \max_j \left[\frac{q_i k_j}{\sqrt{d}} \right] - \frac{1}{L_K} \sum_{j=1}^{L_K} \frac{q_i k_j}{\sqrt{d}}. \quad (8)$$

Note that the n is determined by the top n largest correlation coefficients between feature maps.

- 4) Finally, the frequency components are multiplied with the mask to screen the appropriate frequency components.

C. AED Module for Obtaining FECG Amplitude Components

In the AECG signals, a large number of FECG signals are masked by MECG signals, resulting in a lack of clinical information for fetal diagnosis, as shown in Fig. 1. In addition, the high-amplitude MECG signal with an amplitude of 3–6 times that of the FECG signal (see Fig. 1) will bring the low signal-to-noise FECG signal, further increasing the difficulty of FECG signal extraction. To address such problems, an AED module is devised to obtain the FECG amplitude components masked by MECG, as shown in Fig. 5, which is described as follows.

- 1) First, the amplitude feature maps A_D are first fed into the encoder to generate three feature maps: query, key, and value, where the conv1d layer is employed to extract features and the linear layer play a role in feature encoding.
- 2) Then, these feature maps are passed into the parallel multi-head attention to capturing details from different

representation subspaces [21], [22], where the first one can separate the FECG amplitude from the AECG signals and the second one is used to suppress the noise components

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{SDPA}_1, \dots, \text{SDPA}_n)W_o \quad (9)$$

where W_o denotes the linear matrix, and SDPA is the scaled dot-product attention, defined as

$$\text{SDPA}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (10)$$

in which Softmax denotes the softmax function, d is the scaled factor, and Q , K , and V represent the queries, keys, and values, respectively, calculated from the linear matrixes.

- 3) Finally, the decoder is employed to obtain the FECG amplitude components, where the linear layer can project FECG components into feature maps, and the conv1d layer is utilized to decode the FECG amplitude information.

D. WED Module for Determining FECG Window

Although some hardware facilities are designed for signal filtering, the AECG signal also suffers from various noise sources, as shown in Fig. 1(a). Besides, the patient is typically asked to limit movement and other static measurements in a clinical setting, but the inevitable fetal movement and other interference can cause variations in FECG signals [23]. To solve such problems, some preprocessing measures, such as comb and low-pass filters, are applied to the low-signal-to-noise ratio (SNR) AECG signals, as shown in Fig. 6. However, the FECG signal is weak and affected by the noise, and it is still difficult to locate the FECG signal in the sequence. Motivated by the observations, the WED module is designed to separate the FECG signal corrupted by the noise by determining its temporal window in the sequence, as shown in Fig. 7, which are described as follows.

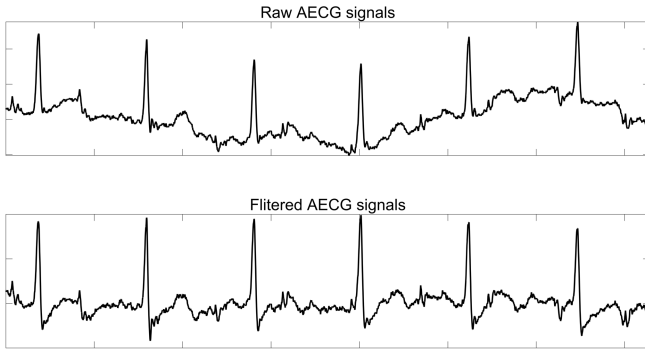


Fig. 6. Examples of filtered AECG signals after comb and low-pass filter processing.

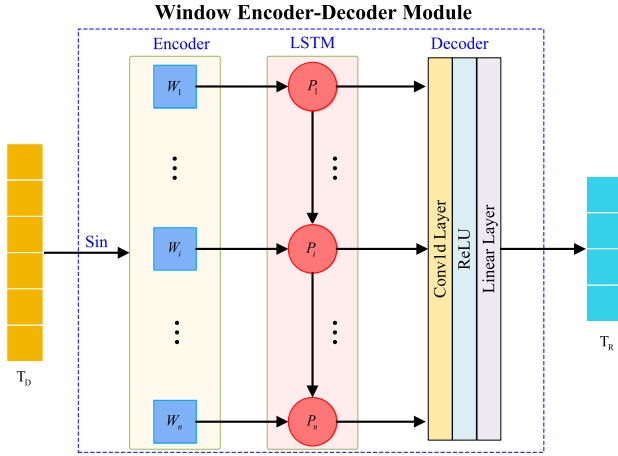


Fig. 7. Structure of WED module. The encoder is first used to perform coordinate embedding of the input sequence. Then, these embedding vectors are fed into the LSTM network for long-term dependency modeling. Finally, the decoder is employed to determine the temporal window in the sequence.

- 1) First, the coordinates of the AECG signal are encoded by

$$E(C) = \text{Sin}\left(C_i * e^{-\frac{2l}{s\sqrt{d}} * l}\right) \quad (11)$$

where Sin is the sine function, C_i denotes the coordinates in the sequence, s represents sampling frequency, and l is the encoding vector dimension.

- 2) Then, the encoder matrix W is applied to establish a dynamic embedding tensor to retain the positional information of FECG signals.
- 3) Next, the LSTM network is exploited to learn the long-term dependency of coordinates in FECG signals, where the LSTM network can search for the coordinates of FECG signals to distinguish them from noise by modeling the periodic dependency in the sequence [24].
- 4) Finally, the encoding information will be fed into the decoder to produce the coordinates weight of FECG signals, where the conv1d layer can obtain the weight of each coordinate, the ReLU function is used to screen the weight, and the linear layer can generate the coordinates weights of the FECG signal in the sequence.

E. Fourier Series Reconstruction

In this part, the frequency, amplitude, and window are utilized to reconstruct the FECG signals, defined as

$$F(t) = \sum_{n=1}^N T_F(t-w) [A_n^F \sin(nwt)] \quad (12)$$

where A_n^F denotes the amplitude, nw represents the frequency, and $T_F(t-w)$ is the window function, expressed by

$$T_F(t-w) = [a_1, \dots, a_n], \quad 0 < a_i < 1. \quad (13)$$

To further improve the robustness of AAFA-Net, the hyperparameters λ_i are introduced to the (12)

$$F(t) = \sum_{n=1}^N T_F(t-w) [A_n^F \sin(nwt)] + \lambda_i \sum_{n=1}^N G_i \quad (14)$$

where G_i denotes the Gaussian white noise.

F. Implementations

During the experiments, the AECG signals are first normalized to $[-1, 1]$ to perform FECG extraction [24]. Then, the Adam optimizer [25] and MSE function [26] are used to train the proposed AAFA-Net on a workstation with four NVIDIA GeForce GTX 1080Ti GPUs, where the batch size, epoch, and learning rate are set as 32, 100, $1e^{-4}$, respectively. In addition, a digital comb filter is applied to eliminate baseline drift [27], and the discrete wavelet transform (DWT) [28] is applied to remove other noise.

III. MATERIALS AND EVALUATION METRICS

A. Dataset

In this article, two FECG datasets, i.e., B2_LABOUR and ADFECGDB datasets, are used as the benchmarks to evaluate the effectiveness of the AAFA-Net, which are described as follows.

B2_LABOUR: This dataset [29] is sampled directly from the fetal head at 1 kHz, and it consists of 12 records about 5 min, while the corresponding MECG signals are sampled from the maternal abdomen at 500 Hz. Besides, the locations of maternal and fetal QRS complexes are provided to researchers. During the experiments, the 12-fold cross-validation is conducted on this dataset, where one recording is used as the validation set, and the rest are regarded as the training set.

ADFECDGB: This dataset [30] consists of five FECG signals sampled at 1 kHz, i.e., r01, r04, r07, r08, and r10. Each signal contains five channel recordings, where the first one is the FECG signal directly collected from the fetal head, and the rest four recordings are the AECG signals. During the experiments, the fivefold cross-validation is used to validate the performance of the proposed AAFA-Net.

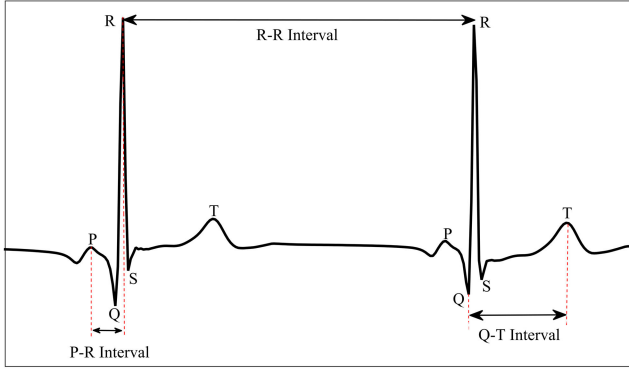


Fig. 8. Clinical metrics of FECG signals, including RR interval, PR interval, and QT interval.

B. Evaluation Metrics

To validate the performance of the AAFA-Net, the mean square error (MSE) [31], the R -squared goodness of fit (R^2) [32], intra-class correlation coefficient (ICC) [33], are exploited to evaluate the quality of FECG signals

$$\text{MSE} = \frac{1}{N} \sum_{n=1}^N [\hat{S}_f(n) - S_f(n)]^2 \quad (15)$$

and

$$R^2 = 1 - \frac{\sum_{n=1}^N [\hat{S}_f(n) - S_f(n)]^2}{\sum_{n=1}^N [\tilde{S}_f(n) - S_f(n)]^2} \quad (16)$$

where $S_f(n)$ represents the actual FECG signal, and $\hat{S}_f(n)$ is the estimated FECG signal.

To verify the clinical practice of the proposed AAFA-Net, three clinical metrics. i.e., R - R , P - R , and Q - T intervals, derived from [34], are tested, as shown in Fig. 8, defined as

$$R-R = T_{R+1} - T_R \quad (17)$$

$$P-R = T_R - T_P \quad (18)$$

and

$$Q-T = T_T - T_Q \quad (19)$$

where T_R , T_P , and T_T are the locations of R -peak, P -peak, and T -peak, respectively, and T_Q denotes the coordinate of Q complex.

IV. EXPERIMENTS AND RESULTS ANALYSIS

A. FECG Signal Quality Evaluation

To verify the performance of AAFA-Net, we compare it with four methods on ADFECGDB and B2_LABOUR datasets.

1) *Comparisons on ADFECGDB Dataset:* We first evaluate the performance of the AAFA-Net on the ADFECGDB dataset using Behar et al. [35], Varanini et al. [36], encoder-decoder [37], and CycleGAN [15]. The results are listed in Table II. It can be seen that the AAFA-Net shows the best performance on FECG extraction, achieving the MSE of 6.31, ICC of 97.92%, and R^2 of 98.83%, respectively, demonstrating the AAFA-Net is an effective method for FECG signals separation. In addition, compared with traditional methods, i.e., [35]

TABLE II
PERFORMANCE RESULTS OF DIFFERENT METHODS FOR FECG
EXTRACTION ON THE ADFECGDB AND
B2_LABOUR DATASETS

Datasets	Methods	MSE	ICC (%)	R^2 (%)
ADFECDGB	Behar et al. [35]	13.41±0.00	91.29±0.00	90.77±0.00
	Varanini et al. [36]	11.38±0.00	92.54±0.00	91.85±0.00
	Encoder-Decoder [37]	8.61±0.24	94.06±0.12	95.33±0.32
	CycleGAN [15]	7.44±0.13	96.76±0.04	97.91±0.02
	AAFA-Net	6.31±0.29	97.92±0.05	98.83±0.06
B2_LABOUR	Behar et al. [35]	15.23±0.00	88.91±0.00	89.71±0.00
	Varanini et al. [36]	14.02±0.00	91.27±0.00	92.02±0.00
	Encoder-Decoder [37]	9.39±0.07	93.56±0.09	94.57±0.05
	CycleGAN [15]	8.06±0.18	96.11±0.12	95.97±0.07
	AAFA-Net	6.87±0.07	98.01±0.11	97.88±0.11

and [36], the AAFA-Net has made substantial advancement with the MSE of 7.10, ICC of 6.63%, and R^2 of 8.06% than Behar et al. [35], showing that it is more sensitive to FECG signals. Moreover, the AAFA-Net outperforms the deep learning methods, i.e., [15] and [37], indicating its powerful ability to feature encoder-decoder on FECG signals.

2) *Comparisons on B2_LABOUR Dataset:* Then, we explore the effectiveness of AAFA-Net on the B2_LABOUR dataset using the same four approaches. The results are shown in Table II, and we can observe that the AAFA-Net can achieve excellent performance in FECG signal separation, with the MSE of 6.87, ICC of 98.01%, and R^2 of 97.88%, respectively, indicating that the robustness of AAFA-Net to fetal complexes is stronger than them. Similarly, the AAFA-Net has achieved substantial improvements of 9.10% and 8.17% on ICC and R^2 , respectively, compared to Behar et al. [35], showing that it has the advantage of FECG extraction based on frequency, amplitude, and window spaces. In addition, it can be noted that the AAFA-Net has achieved slight improvements over the CycleGAN [15], with the MSE of 1.19, ICC of 1.90%, and R^2 of 2.11%, respectively, demonstrating that the attention to amplitude and frequency can perform better than the attention to FECG waveforms.

B. Clinical Practice Assessment

To verify the practicality of the proposed AAFA-Net in the clinic, three clinic metrics, i.e., R - R interval, P - R interval, and Q - T interval are used for evaluation on the ADFECGDB dataset using Varanini et al. [36], encoder-decoder [37], and CycleGAN [15]. Five cross-validation experiments are conducted on this dataset. Table III lists the results, and the boxplots are shown in Fig. 9, where labels denote the FECG signals collected from the fetal head. Here, the Pan and Tompkins [38] and NeuroKit [39] are applied to detect the P , Q , R , S , and T waves in the predicted FECG signals and labels.

It can be inferred from Table III that the AAFA-Net can obtain the best results on R - R calculation, achieving 127.41, 125.83, 126.07, 134.14, and 118.51 bpm on $r01$, $r04$, $r07$, $r08$, and $r10$ recordings, respectively, which are closest to the actual ones. This demonstrates that the AAFA-Net is reliable in fetal heart rate estimation, further assisting obstetricians in

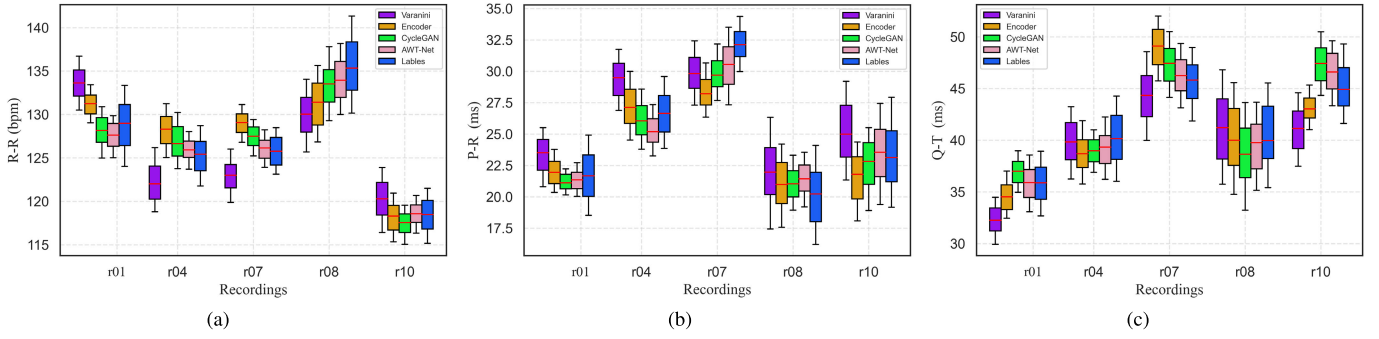


Fig. 9. Boxplots of clinical metrics of different methods on the ADFECGDB dataset. (a) RR interval. (b) PR interval. (c) QT interval.

TABLE III
R-R, P-R, AND Q-T INTERVALS OF DIFFERENT
METHODS ON THE ADFECGDB DATASET

Recordings	Methods	R-R (bpm)	P-R (ms)	Q-T (ms)
r01	Varanini <i>et al.</i> [36]	133.67±3.19	23.17±2.37	32.21±2.29
	Encoder-Decoder [37]	131.29±2.24	22.06±1.72	34.78±2.32
	CycleGAN [15]	127.94±3.01	21.19±1.04	36.97±2.02
	AAFA-Net	127.41±2.49	21.37±1.35	35.83±2.76
	Labels	128.67±4.73	21.72±3.22	35.85±3.17
r04	Varanini <i>et al.</i> [36]	122.51±3.71	29.33±2.46	39.72±3.52
	Encoder-Decoder [37]	128.14±3.11	27.26±2.74	38.91±3.15
	CycleGAN [15]	126.96±3.28	26.17±2.42	38.97±2.08
	AAFA-Net	125.83±2.21	25.31±2.06	39.28±3.07
	Labels	125.26±3.51	26.72±2.91	40.14±4.17
r07	Varanini <i>et al.</i> [36]	122.95±3.09	29.87±2.57	44.32±4.35
	Encoder-Decoder [37]	128.92±2.26	28.51±2.17	48.83±3.19
	CycleGAN [15]	127.36±2.13	29.93±2.26	47.35±3.21
	AAFA-Net	126.07±2.19	30.41±3.11	46.28±3.16
	Labels	125.78±2.73	32.17±2.22	45.38±3.59
r08	Varanini <i>et al.</i> [36]	129.88±4.19	21.91±4.47	41.29±5.59
	Encoder-Decoder [37]	131.26±4.41	20.93±3.37	40.18±5.41
	CycleGAN [15]	133.54±4.27	21.12±2.21	38.43±5.22
	AAFA-Net	134.14±4.15	21.38±2.18	39.42±4.27
	Labels	135.74±5.63	20.13±3.97	40.46±5.11
r10	Varanini <i>et al.</i> [36]	120.17±3.77	25.32±4.01	41.08±3.61
	Encoder-Decoder [37]	118.14±2.81	21.23±3.15	43.21±2.19
	CycleGAN [15]	117.26±2.29	22.22±3.32	47.43±3.11
	AAFA-Net	118.51±2.18	23.42±4.03	46.51±3.16
	Labels	118.34±3.19	23.56±4.42	45.49±3.87

diagnosing fetal well-being. In addition, the P - R intervals are also listed in Table III and Fig. 9(b), which can provide quantitative references for experts to diagnose heart disease [40]. Although the AAFA-Net cannot achieve the optimal value, the error between it and the actual label is in a small range, about 0.35, 1.41, 1.76, 1.25, and 0.14 ms on $r01$, $r04$, $r07$, $r08$, and $r10$ recordings, respectively, which is still beneficial for fetal diagnosis.

Moreover, it is necessary to monitor fetal hypoxia, which is contained in the ST -segment of FECG signals [7]. For example, the rising T wave often indicates a fetal response to hypoxia [41]. To this end, the Q - T interval is estimated from the FECG signals. It can be seen from Fig. 9(c) that the Q - T intervals generated by the proposed AAFA-Net are most consistent with the actual ones, about 35.83, 39.28, 46.28, 39.42, and 46.51 ms, where the labels are 35.85, 40.14, 45.38, 40.46, and 45.49 ms on the $r01$, $r04$, $r07$, $r08$,

TABLE IV
ABLATION STUDIES FOR THE PROPOSED MODULES IN AAFA-NET
TESTED ON THE ADFECGDB DATABASE

No.	Baseline	FED	AED	WED	ATT	MSE	ICC (%)	R^2 (%)
1	✓					8.66±0.31	94.19±0.22	95.21±0.18
2	✓	✓	✓			7.01±0.09	97.02±0.25	97.25±0.17
3	✓	✓	✓	✓		6.58±0.14	97.73±0.13	98.11±0.19
4	✓	✓	✓	✓	✓	6.16±0.11	98.03±0.21	98.89±0.27

and $r10$ recordings, respectively. Finally, we can also obtain clinical metrics (e.g., ST segment) based on RR , PR , and QT intervals, thereby performing other abnormal diagnoses.

C. Ablation Studies

Here, we have conducted a series of ablation experiments to validate the impacts of the proposed FED module, AED module, and WED module, attention mechanism, top n in correlation sparse attention, and different attention network in AAFA-Net.

1) *Impact of Frequency and AED Module:* In the proposed AAFA-Net, the FED and AED modules are employed to obtain the frequency and amplitude of FECG signals. To verify their effectiveness, a series of ablation experiments are conducted on the ADFECGDB dataset, and the results are shown in Table IV. No.1 is the baseline performance. It can be seen that the baseline can achieve 8.66, 94.19%, and 95.21% for MSE, ICC, and R^2 , respectively. After applying the FED and AED modules to the baseline, the results can be up to 97.02% and 97.25% on the ICC and R^2 , respectively, and the MSE is reduced to 7.01, showing that the FED and AED modules are beneficial for FECG separation. In addition, it can be observed from (b) and (c) in Fig. 1 of Supplementary Information that the FED and AED modules can preserve the FECG signal kurtosis. More specifically, by comparing the FECG signals mixed with MECG signals, i.e., the second, fourth, and fifth ones in Fig. 1(b) and (c) of Supplementary Information, the proposed FED and AED modules are prone to extract FECG signals masked by MECG signals.

2) *Impact of WED Module:* To validate the effect of the WED module, we compare it with the Baseline+FED+AED, and the results are given in Table IV. From the second and third rows in Table IV, adding WED can achieve improvements of 0.71% on ICC, 0.86% on R^2 , and the decrement

TABLE V

RESULTS OF DIFFERENT TOP n IN CORRELATION SPARSE ATTENTION TESTED ON THE ADFECGDB DATASET

Top- n	MSE	ICC (%)	R^2 (%)
6	14.77±0.81	90.24±0.74	89.83±0.69
12	8.19±0.29	95.93±0.28	95.88±0.22
19	6.57±0.15	97.61±0.14	97.63±0.18
22	6.16±0.11	98.03±0.21	98.89±0.27
25	6.69±0.17	97.11±0.25	96.95±0.22
27	7.12±0.21	96.36±0.23	95.37±0.28
28	7.19±0.22	96.22±0.26	95.23±0.27
30	7.51±0.25	95.83±0.28	94.78±0.23

TABLE VI

RESULTS OF DIFFERENT ATTENTION NETWORKS FOR FECG EXTRACTION ON THE ADFECGDB DATASET

Methods	MSE	ICC (%)	R^2 (%)
CA [42]	10.32±0.24	93.37±0.41	94.28±0.29
SE [43]	8.09±0.19	96.61±0.33	97.42±0.26
CS [44]	7.88±0.14	96.92±0.26	97.82±0.22
MH [45]	6.16±0.11	98.03±0.21	98.89±0.27

of 0.43 on MSE. This is because the WED module can adaptively determine the temporal window of FECG signals in the sequence, further assisting the network in distinguishing the FECG signals from noise. Similarly, it can be inferred from (d) and (c) in Fig. 1 of Supplementary Information that the estimated FECG signal in (d) is smoother than that of (c), where the residual noise is adaptively eliminated by the window function, indicating that the WED module can help AAFA-Net to separate FECG signals corrupted by noise.

3) *Impact of Attention Mechanism:* In the proposed FED and AED modules, the attention mechanism is exploited to capture the details of FECG signals. To valid the effect of the attention mechanism, a number of experiments are performed on the ADFECGDB database, and the results are shown in Table IV. Comparing the last row with the third one, we can infer that applying attention to the network can make improvements of 0.3% and 0.78% on ICC and R^2 , respectively, demonstrating the attention mechanism is beneficial for FECG extraction. Furthermore, the visualization result after adding attention is shown in Fig. 1(e) of Supplementary Information. We can see that the predicted FECG signals are almost close to the actual ones, especially in the P - R segment and T wave, showing that the attention mechanism can improve the waveform quality of FECG signals.

4) *Choice of Top n in Correlation Sparse Attention:* In this part, a number of experiments are conducted to explore the impact of top n in correlation sparse attention, where the length of all frequency masks is 32, and the results are recorded in Table V. It can be found that the performance of AAFA-Net is increased with increasing n in correlation sparse attention, and it can achieve the best performance when n is 22, but further increase in n will lead to performance degradation. Therefore, the top n is set to 22 in correlation sparse attention.

5) *Impact of Different Attention Networks:* To investigate the effect of different attention networks for FECG

TABLE VII

RESULTS OF AAFA-NET IN DIFFERENT SNR ON THE ADFECGDB DATASET

SNR	MSE	ICC (%)	R^2 (%)
raw	6.16±0.11	98.03±0.21	98.89±0.27
SNR-5	20.37±2.27	86.19±1.81	86.33±2.19
SNR00	15.29±1.48	90.17±2.62	90.85±2.02
SNR10	10.82±1.07	93.77±1.56	94.13±1.38
SNR20	7.42±0.55	97.49±0.59	97.91±0.42

Extraction, four different attention networks, i.e., channel attention network [42] (CA), channel and spatial attention [44] (CS) network, squeeze and excitation attention [43] (SE) network, and multi-head attention [45] (MH) network, are applied to the proposed AAFA-Net, and the results are given in Table VI. From it we can observe that the multi-head attention network shows the best performance on the ADFECGDB dataset, achieving the results of 6.16, 98.03%, and 98.89% on MSE, ICC, and R^2 , respectively, validating that the multi-head attention network is helpful for FECG signal extraction.

V. DISCUSSION

In this article, an AAFA-Net is proposed for FECG extraction, and the results on the B2_LABOUR and ADFECGDB datasets demonstrate the effectiveness of the proposed AAFA-Net. Anti-noise ability, training and inference time, and DWT for noise reduction are discussed to verify the superiority of AAFA-Net, and the limitations and drawbacks are described.

First, a series of experiments are performed on the ADFECGDB dataset to verify the anti-noise ability of AAFA-Net for FECG extraction at four different SNR, and the SNR is calculated by

$$\text{SNR} = 10 \log_{10} \left(\frac{P_f}{P_n} \right) \quad (20)$$

where P_f and P_n denote the powers of the ECG and the Gaussian noise, defined as

$$P_f = \frac{1}{N} \sum_{i=1}^N f_i^2 \quad (21)$$

$$P_n = \frac{1}{N} \sum_{i=1}^N n_i^2 \quad (22)$$

in which f_i represents the FECG signal, n_i denotes the noise, and N is the length of FECG signals and noise. The results are listed in Tables VII, where the raw represents the original AECG signals, the SNR-5, SNR00, SNR10, and SNR20 represent the SNR at -5, 0, 10, and 20 dB, respectively, as shown in Fig. 2 of Supplementary Information. It can be seen from Table VII that the performance of the model declines with the reduction of the SNR. Specifically, the AAFA-Net shows good results at the SNR of 10 and 20 dB, achieving the R^2 of 94.13% and 97.91%, respectively. However, the MSE is large when the SNR is set to -5 dB, where the FECG is completely

submerged by Gaussian noise. At this time, it is essential to perform preprocessing filtering for the raw abdominal signals.

Besides, to explore the time complexity of the proposed AAFA-Net, we have also given the training and inference time of AAFA-Net, where the training is conducted on the workstation with four NVIDIA GeForce GTX 1080Ti GPUs, and the testing is carried out on a computer with Intel Core i5-6300U and another low-power Raspberry Pi 4B, respectively. Specifically, the training time is 235.89 s on the workstation, and the test times are 4.19 and 52.92 s on the computer and Raspberry Pi 4B, respectively. From the test time on the low-power Raspberry Pi 4B, we can see that the AAFA-Net shows a lower speed for FECG extraction, but it can be a real-time tool on the computer with Intel Core i5-6300U. Therefore, the proposed AAFA-Net can be suitable for low-power hardware with powerful high-speed processors.

In addition, it can be seen from Table III that there is a minuscule difference in the performance between AAFA-Net and CycleGAN. To show the superiority of the proposed AAFA-Net, we have tested the two models on the computer with Intel Core i5-6300U in terms of performance and computational complexity. Specifically, the FPS, FLOPs, and Params of CycleGAN are 18.28, 11.51 M, and 76.44 K, respectively, while the ones of AAFA-Net are 87.92, 17.79 M, 482.48 K, respectively. From these results, we can observe that the proposed AAFA-Net is 4.8 times faster than CycleGAN about the FPS even though AAFA-Net has more FLOPs than that of CycleGAN. This is because the AAFA-Net relies more on linear layers than convolution layers, as shown in Table I. Moreover, it can be found from Tables II and III that the proposed AAFA-Net has shown better performance than CycleGAN, further providing more reliable results and clearer FECG signal morphology for clinical diagnosis. By comparing performance and computational complexity, the proposed AAFA-Net is more suitable for clinical settings.

Furthermore, the application of DWT in FECG signals denoising is discussed, and an example of the DWT for AECG signals is shown in Fig. 3 of Supplementary Information. From it, we can note that the FECG signals (marked by the red circle) become clearer and the noise is gradually eliminated. These results demonstrate that the DWT can be used for noise elimination in FECG extraction, and further improve the performance of the proposed AAFA-Net. Besides, the P and T waves are apparent at the fourth layer, while they are invisible on previous layers. Therefore, the four-layer DWT is applied to remove the noise from the FECG signals.

Finally, the limitations and drawbacks are discussed. From the above experiments, the proposed AAFA-Net shows the best performance on two FECG datasets compared to other methods. However, there are still several limitations as follows. First, although the proposed AAFA-Net shows excellent anti-noise ability for FECG extraction, it is still necessary to perform preprocessing filtering to further improve the anti-noise capability at the low SNR, such as FECG signals submerged by noise. Second, the proposed AAFA-Net is tested on two limited FECG datasets, it should be conducted on larger datasets to improve the diagnosis performance. Third, these two datasets only contain a small number of fetal arrhythmic

events and ectopic beats, leading to their limitations in clinical practice. To this end, a large dataset covering relevant arrhythmic situations and ectopic beats should be collected to address this problem in future works.

VI. CONCLUSION

In this article, an AAFA-Net is proposed for FECG separation, where the FED module is devised to discriminate the FECG frequency components from MECG signals, and the AED module is designed to extract the FECG amplitude components masked by MECG signals, while the WED module is developed to separate the FECG signals corrupted by the noise by determining the temporal window in the sequence. The results show that the AAFA-Net outperforms the state-of-the-art methods, achieving the ICC of 97.92% and 98.01% on ADFECGDB and B2_LABOUR datasets, respectively, further assisting obstetricians in performing fetal diagnosis.

REFERENCES

- [1] S. Robbani et al., "An implantable electronic device for monitoring fetal lung pressure in a Lamb model of congenital diaphragmatic hernia," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–10, 2022.
- [2] J. Zhang, Z. L. Yu, Z. Gu, Y. Li, and Z. Lin, "Multichannel electrocardiogram reconstruction in wireless body sensor networks through weighted $\ell_{1,2}$ minimization," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 9, pp. 2024–2034, Sep. 2018.
- [3] R. Avci, J. D. Wilson, D. Escalona-Vargas, and H. Eswaran, "Tracking fetal movement through source localization from multisensor magnetocardiographic recordings," *IEEE J. Biomed. Health Informat.*, vol. 22, no. 3, pp. 758–765, May 2018.
- [4] P. Hamelmann, R. Vullings, M. Mischi, A. F. Kolen, L. Schmitt, and J. W. M. Bergmans, "An extended Kalman filter for fetal heart location estimation during Doppler-based heart rate monitoring," *IEEE Trans. Instrum. Meas.*, vol. 68, no. 9, pp. 3221–3231, Sep. 2019.
- [5] P. Hamelmann et al., "Doppler ultrasound technology for fetal heart rate monitoring: A review," *IEEE Trans. Ultrason., Ferroelectr., Freq. Control*, vol. 67, no. 2, pp. 226–238, Feb. 2020.
- [6] J. Wang, K. Yang, R. Yang, X. Kong, and W. Chen, "Squid gradiometer module for fetal magnetocardiography measurements inside a thin magnetically shielded room," *IEEE Trans. Appl. Supercond.*, vol. 29, no. 2, pp. 1–4, Mar. 2019.
- [7] C. Lin et al., "Robust fetal heart beat detection via R-peak intervals distribution," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 12, pp. 3310–3319, Dec. 2019.
- [8] G. J. J. Warmerdam, R. Vullings, L. Schmitt, J. O. E. H. Van Laar, and J. W. M. Bergmans, "Hierarchical probabilistic framework for fetal R-Peak detection, using ECG waveform and heart rate information," *IEEE Trans. Signal Process.*, vol. 66, no. 16, pp. 4388–4397, Aug. 2018.
- [9] D. P. Tobón and T. H. Falk, "Adaptive spectro-temporal filtering for electrocardiogram signal enhancement," *IEEE J. Biomed. Health Informat.*, vol. 22, no. 2, pp. 421–428, Mar. 2018.
- [10] L. Wang, T. Ohtsuki, K. Owada, N. Honma, and H. Hayashi, "A multi-layer hybrid network with its application in fetal heart rate monitoring," *IEEE Signal Process. Lett.*, vol. 29, pp. 1207–1211, 2022.
- [11] W.-L. Hwang and J. Ho, "Null space component analysis of one-shot single-channel source separation problem," *IEEE Trans. Signal Process.*, vol. 69, pp. 2233–2251, 2021.
- [12] D. Gurve and S. Krishnan, "Separation of fetal-ECG from single-channel abdominal ECG using activation scaled non-negative matrix factorization," *IEEE J. Biomed. Health Informat.*, vol. 24, no. 3, pp. 669–680, Mar. 2020.
- [13] M. Boussé, O. Debals, and L. De Lathauwer, "A tensor-based method for large-scale blind source separation using segmentation," *IEEE Trans. Signal Process.*, vol. 65, no. 2, pp. 346–358, Jan. 2017.
- [14] O. Debals, M. Van Barel, and L. De Lathauwer, "Löwner-based blind signal separation of rational functions with applications," *IEEE Trans. Signal Process.*, vol. 64, no. 8, pp. 1909–1918, Apr. 2016.

- [15] M. R. Mohebbian, S. S. Vedaiei, K. A. Wahid, A. Dinh, H. R. Marateb, and K. Tavakolian, "Fetal ECG extraction from maternal ECG using attention-based CycleGAN," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 2, pp. 515–526, Feb. 2022.
- [16] F. Andreotti, F. Gräßer, H. Malberg, and S. Zaunseder, "Non-invasive fetal ECG signal quality assessment for multichannel heart rate estimation," *IEEE Trans. Biomed. Eng.*, vol. 64, no. 12, pp. 2793–2802, Dec. 2017.
- [17] H. Zhang, G. Wang, P. Cai, Z. Wu, and S. Ding, "A fast blind source separation algorithm based on the temporal structure of signals," *Neurocomputing*, vol. 139, pp. 261–271, Sep. 2014.
- [18] Z. Yang, W. Yan, X. Huang, and L. Mei, "Adaptive temporal-frequency network for time-series forecasting," *IEEE Trans. Knowl. Data Eng.*, vol. 34, no. 4, pp. 1576–1587, Apr. 2022.
- [19] C. Tan, L. Zhang, and H.-t. Wu, "A novel Blaschke unwinding adaptive-Fourier-decomposition-based signal compression algorithm with application on ECG signals," *IEEE J. Biomed. Health Informat.*, vol. 23, no. 2, pp. 672–682, Mar. 2019.
- [20] H. Zhou et al., "Informer: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, no. 12, 2021, pp. 11106–11115.
- [21] Y. Deng, X. Wang, and Y. Liao, "ASA-Net: Adaptive sparse attention network for robust electric load forecasting," *IEEE Internet Things J.*, early access, Aug. 1, 2023, doi: [10.1109/JIOT.2023.3300695](https://doi.org/10.1109/JIOT.2023.3300695).
- [22] X. Wang et al., "CGA-UNet: Category-guide attention U-Net for dental abnormality detection and segmentation from dental-maxillofacial images," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–11, 2023.
- [23] R. Vullings, M. Mischi, S. G. Oei, and J. W. M. Bergmans, "Novel Bayesian vectorcardiographic loop alignment for improved monitoring of ECG and fetal movement," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 6, pp. 1580–1588, Jun. 2013.
- [24] X. Wang et al., "PA²Net: Period-aware attention network for robust fetal ECG detection," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–12, 2022.
- [25] D. P. Kingma and J. Ba, "ADAM: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2015, pp. 1–15.
- [26] S. Jadon, "A survey of loss functions for semantic segmentation," in *Proc. IEEE Conf. Comput. Intell. Bioinf. Comput. Biol. (CIBCB)*, Oct. 2020, pp. 1–7.
- [27] H. Wang, Z. Zhao, Z. Wang, G. Xu, and L. Wang, "Independent component analysis-based baseline drift interference suppression of portable spectrometer for optical electronic nose of Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 16, no. 4, pp. 2698–2706, Apr. 2020.
- [28] N. Holighaus, G. Koliander, Z. Prusa, and L. D. Abreu, "Characterization of analytic wavelet transforms and a new phaseless reconstruction algorithm," *IEEE Trans. Signal Process.*, vol. 67, no. 15, pp. 3894–3908, Aug. 2019.
- [29] I. Silva et al., "Noninvasive fetal ECG: The PhysioNet/computing in cardiology challenge 2013," in *Proc. Comput. Cardiol.*, Sep. 2013, pp. 149–152.
- [30] J. Jezewski, A. Matonia, T. Kupka, D. Roj, and R. Czabanski, "Determination of fetal heart rate from abdominal signals: Evaluation of beat-to-beat accuracy in relation to the direct fetal electrocardiogram," *Biomedizinische Technik/Biomed. Eng.*, vol. 57, no. 5, pp. 383–394, Jan. 2012.
- [31] M. Alfaouri and K. Daqrouq, "Quality evaluation techniques of processing the ECG signal," *Amer. J. Appl. Sci.*, vol. 5, no. 12, pp. 1737–1741, Dec. 2008.
- [32] J. Miles, *R Squared, Adjusted R Squared*. Hoboken, NJ, USA: Wiley, 2014.
- [33] S. Mehta et al., "Performance of intraclass correlation coefficient (ICC) as a reliability index under various distributions in scale reliability studies," *Statist. Med.*, vol. 37, no. 18, pp. 2734–2752, Aug. 2018.
- [34] D. J. Jagannath and A. I. Selvakumar, "Superior fetal electrocardiogram signal elicitation using a novel artificial intelligent Bayesian methodology," *Appl. Soft Comput.*, vol. 37, pp. 1002–1017, Dec. 2015.
- [35] J. Behar, A. Johnson, G. D. Clifford, and J. Oster, "A comparison of single channel fetal ECG extraction methods," *Ann. Biomed. Eng.*, vol. 42, no. 6, pp. 1340–1353, Jun. 2014.
- [36] M. Varanini, G. Tartarisco, L. Billeci, A. Macerata, G. Pioggia, and R. Balocchi, "An efficient unsupervised fetal QRS complex detection from abdominal maternal ECG," *Physiol. Meas.*, vol. 35, no. 8, pp. 1607–1619, Aug. 2014.
- [37] W. Zhong, L. Liao, X. Guo, and G. Wang, "Fetal electrocardiography extraction with residual convolutional encoder-decoder networks," *Australas. Phys. Eng. Sci. Med.*, vol. 42, no. 4, pp. 1081–1089, Dec. 2019.
- [38] J. Pan and W. J. Tompkins, "A real-time QRS detection algorithm," *IEEE Trans. Biomed. Eng.*, vol. BME-32, no. 3, pp. 230–236, Mar. 1985.
- [39] D. Makowski et al., "NeuroKit2: A Python toolbox for neurophysiological signal processing," *Behav. Res. Methods*, vol. 53, no. 4, pp. 1689–1696, Aug. 2021.
- [40] A. J. Rogers, J. M. Miller, R. Kannappan, and P. Sethu, "Cardiac tissue chips (CTCs) for modeling cardiovascular disease," *IEEE Trans. Biomed. Eng.*, vol. 66, no. 12, pp. 3436–3443, Dec. 2019.
- [41] M. L. Cabaniss and M. G. Ross, *Fetal Monitoring Interpretation*. Philadelphia, PA, USA: Lippincott Williams & Wilkins, 2010.
- [42] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 3–19.
- [43] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
- [44] A. He, T. Li, N. Li, K. Wang, and H. Fu, "CABNet: Category attention block for imbalanced diabetic retinopathy grading," *IEEE Trans. Med. Imag.*, vol. 40, no. 1, pp. 143–153, Jan. 2021.
- [45] A. Vaswani et al., "Attention is all you need," in *Proc. Int. Conf. Neural Inf. Process. Syst. (NIPS)*, 2017, pp. 5998–6008.



Yong Liao was born in Loudi, Hunan, China, in 1990. He received the Ph.D. degree from the Guangdong University of Technology, Guangzhou, China, in 2021.

Now, he is an Assistant Professor at the School of Physics, Electronics and Electrical Engineering, Xiangnan University, Chenzhou, China. His research interests include signal processing, intelligent manufacturing system design and optimization.



Xu Wang was born in Hunan, China. He received the B.S. degree in electrical engineering and automation from Xiangnan University, Chenzhou, China, in 2014, and the M.S. degree in control science and engineering from the Guangdong University of Technology, Guangzhou, China, in 2017, where he is currently pursuing the Ph.D. degree in control science and engineering.

His research interests include biomedical signal processing and deep learning.



Yang Han received the B.S. degree in clinical specialty from Sun Yat-sen University, Guangzhou, China, in 2011, and the M.S. degree in gynecology and obstetrics from Sun Yat-sen Memorial Hospital, Guangzhou, in 2014. She is currently pursuing the Ph.D. degree with First Affiliated Hospital, Sun Yat-sen University, Guangzhou.

Her research interests include obstetric monitoring and diagnosis.



Yamei Deng received the B.S. degree in medical image science from Xiangnan University, Chenzhou, China, in 2015.

She is currently with the Department of Radiology, The Third Affiliated Hospital of Guangzhou University of Medical, Guangzhou, China. Her research interests include medical image in fetal monitoring and diagnosis.